Headline XV: the assembled leading posterior quotient

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Abstract

Unit 208 closes programme B2 by feeding the conditional assembly theorem of unit 207 with the phase-dressed chart integrals of Headlines XIII–XIV. For a finite family of charts $a \in \iota$, each with its own dimension, exponents (h_a, k_a), minimum ratio λ_a ($p_a = 2\lambda_a$), multiplicity m_a, continuous phase ξ_a, observable φ_a and density $c_a > 0$ on the closed cube, all at the same β,

$$\frac{\sum_a \int_{(0,1]^{d_a}} \varphi_a c_a\, u^{h_a} e^{-\beta N^2 u^{2k_a} + \beta N u^{k_a} \xi_a}}{\sum_a \int_{(0,1]^{d_a}} c_a\, u^{h_a} e^{-\beta N^2 u^{2k_a} + \beta N u^{k_a} \xi_a}} \longrightarrow \frac{\sum_{a \in \mathrm{sel}} \mathrm{phaseCoeff}_a[\varphi_a c_a]}{\sum_{a \in \mathrm{sel}} \mathrm{phaseCoeff}_a[c_a]},$$

where sel is the set of charts with $p_a = \min_b p_b$ and $m_a = \max\{m_b : p_b = \min\}$ (`headline_assembled_posterior`). The denominator is positive since each selected $\mathrm{phaseCoeff}_a[c_a] > 0$ and the selection is nonempty. Zero `sorry`/`axiom`; 9085 jobs.

1 The things formalised

```
noncomputable def chartIntegral (d) (h k : Fin d →  ) (  N :  ) (   ) :    :=
   u in unitBox d,   u * ((  i, u i ^ h i) * quadKernel   (  u) (N *   i, u i ^ k
     i))
theorem chartIntegral_eq : chartIntegral d h k   N
    =   u in unitBox d,   u * ((  u^h) * exp (-(  N²   u^(2k)) +   (N   u^k)   u))
theorem headline_assembled_posterior (d :   →  ) (h k : (a :  ) → Fin (d a + 1) →
      ) (hk) (l :   →  )
    ( ) (hl) (h ) (hmin) (hatt) (    c : (a :  ) → (Fin (d a + 1) →  ) →  ) (h )
      (h ) (hc)
    (hcpos :   a,   x   closedCube (d a + 1), 0 < c a x) :
    Tendsto (fun N => (  a, chartIntegral (d a + 1) (h a) (k a)   N (  a) (fun x =>
       a x * c a x))
       /   a, chartIntegral (d a + 1) (h a) (k a)   N (  a) (c a)) atTop
      (  ((  a   univ.filter (sel), phaseCoeff (h a) (k a) (l a)   (  a) (fun x =>
       a x * c a x))
         /   a   univ.filter (sel), phaseCoeff (h a) (k a) (l a)   (  a) (c a)))
  -- sel a := 2 l a = leadExp (fun b => 2 l b)   multCount (ratioExp (h a) (k a))
      (l a)
  --                 = leadMult (fun b => 2 l b) (fun b => multCount (ratioExp (h b)
      (k b)) (l b))
```

2 Mathematics and scope

Nothing new analytically: `assembly_ratio_tendsto` with I_a the chart integrals, $p_a = 2\lambda_a$, $m_a =$ `multCount`, limits from `phase_leading_tendsto`, and positivity of the selected denominator sum from `phaseCoeff_pos` and `exists_leadMult`. Dependent types carry the per-chart dimensions (`h k : (a :  ) → Fin (d a + 1) →  `).

What the theorem says: *given* that the posterior numerator and denominator are these finite sums of chart integrals, the leading quotient is the selected-chart face average. What it does not say: that the resolution, Jacobians, partition of unity and tangential integrations produce such a sum (the paper-level input), nor anything about random phases. Astra #22's gates are met: minimum exponent then *maximum* multiplicity; numerator and denominator assembled separately; denominator positivity via an explicit interface (here $c_a > 0$; Astra suggested allowing vanishing partition amplitudes with positivity of the assembled coefficient instead — a one-line weakening of the hypothesis `hpos` in unit 207's `assembly_ratio_tendsto`, already available).

3 Lean notes

- A family of charts with varying dimension is a dependent function `h : (a :  ) → Fin (d a + 1) →  `; all downstream lemmas apply pointwise in `a` without friction.
- The selection predicate must be spelled out identically in the statement and in the `Finset.filter` of `assembly_ratio_tendsto` for `exact` to unify; using `fun b => 2 * l b` for `p` and `fun b => multCount …` for `m` throughout makes this syntactic.

4 Status

B2 complete (units 207–208). Remaining per Astra #22: A2 (general-d stochastic transfer via equi-Lipschitz control), D2 (signed orthants with phase), hand-off.